Sahaj Bhadja

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Summary

Software engineer with 2+ years building **backend services**, **secure APIs**, and **cloud-backed data workflows**. Strong in Java, Python, AWS, PostgreSQL, CI/CD, automated testing, and AI-assisted development for reliable production systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected Azure SQL and Blob-backed services for 500+ users, tuning indexed queries and access patterns to cut dashboard latency from 1.2 seconds to 350 milliseconds across internal analytics workflows.
- Built fault-tolerant job execution for 1,000+ monthly workflows with explicit state transitions, retries, and isolation, improving throughput by 3x while strengthening backend observability, debugging, and production reliability under load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Flask APIs and instructor-facing workflows handling 10,000+ daily requests, integrating secure file ingestion and AI-assisted grading operations to deliver maintainable backend services and supportable full-stack product features.
- Designed multithreaded processing pipelines for 1,000+ uploads on AWS RDS-backed services, reducing feedback turnaround by 60% while improving throughput, deployment confidence, and operational stability for concurrent production workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built evaluation pipelines over 100K+ trajectory logs for transformer-based behavioral planning, improving long-horizon trajectory quality by 18% in simulation while strengthening large-scale data processing and experiment reliability.
- Architected a spatial-temporal encoder combining HD-map geometry with agent history through multi-head attention, improving closed-loop planning stability by 15% and sharpening analytical debugging across complex model behaviors.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered Python-based API and ETL microservices with AWS Glue, S3, and SQL integrations, improving response times by 30% while supporting reusable backend services, low-bandwidth clients, and maintainable enterprise workflows.
- Maintained Jenkins CI/CD pipelines and 20+ automated suites with PyTest and Mockito, increasing release reliability by 68% while strengthening regression coverage, cross-platform validation, and secure production deployment practices.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built full-stack MERN APIs and WebSocket modules supporting 2M+ synchronized records, improving real-time consistency while keeping eight business services modular, scalable, and easier to maintain across CRM workflows.
- Automated Docker-based CI/CD workflows with Jenkins, staged testing, and deployment validation, cutting release time by 50% while improving troubleshooting speed, build consistency, and delivery reliability for distributed application updates.

Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, Bash, C++; **Backend & APIs**: Flask, FastAPI, Node.js, Express.js, REST APIs, microservices, WebSockets, object-oriented design, unit testing, integration testing
Cloud & Data: AWS, EC2, S3, RDS, AWS Glue, Azure, Azure SQL, PostgreSQL, MySQL, MongoDB, Firebase, SQL Server, ETL pipelines; **Delivery**: Docker, Kubernetes, Jenkins, GitHub Actions, Git, CI/CD, monitoring, observability, Agile, distributed systems, system design, performance optimization, AI-assisted development

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL development workspace for 10+ databases, accelerating query authoring and schema exploration through unified Java and TypeScript services with Docker-backed workflows for maintainable enterprise-style data operations.

Linguistix | *Python, AWS, Docker, CI/CD*
- Developed and deployed a containerized RESTful data processing API on AWS with Docker and automated CI/CD releases, adding validation checks and achieving production-grade reliability with 87% processing accuracy.

Big-Brain Live | *PostgreSQL, FAISS, GCP, REST APIs*
- Built secure document-upload and semantic-retrieval workflows over 50K+ documents using PostgreSQL, FAISS, and REST APIs, reducing review time by 60% while supporting low-latency search and collaborative investigation flows.